Decaying Autoregressive Memory Models for Conditional Volatility:

Mathematical Foundations, Finite-Sample Identification and Out-of-Sample Evidence

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Abstract

Decaying AutoRegressive Memory (DARM) is a normalised, parsimoniously parameterised ARCH(K)/ARCH(∞) conditional-variance specification that separates aggregate volatility feedback from the distribution of lag weights. We give explicit assumptions for causal stationarity, moments, finite-lag renewal-response persistence, power-law weight identification, squared-return spectral representations and multi-step forecasting. These are parameterisation-specific deductions within established ARCH theory rather than a new stochastic-process class. In an independent 2020–2025 evaluation across 75 US equities, asymmetric DARM recorded the lowest mean QLIKE of the fitted benchmarks, but its designated 10-day advantage over Student-t GJR-GARCH was not significant ($\Delta\text{QLIKE} = 0.015048$, HAC $p = 0.146291$, Holm-adjusted $p = 0.292582$). Synthetic Experiments 13–18 show that kernel parameters and implied long-run half-life can be weakly identified despite accurate short-horizon variance forecasts; DARM is competitive under selected ARCH-family generating processes but does not dominate return-filtered stochastic volatility under persistent log-SV. In Experiment 18, near-nominal intervals for future squared returns can coexist with severe undercoverage for latent integrated conditional variance. These findings delimit what variance point-forecast performance implies about structural persistence and latent risk. Original source notebooks and Python implementations for Experiments 13–18 are now archived with reported outputs and validation tools; nevertheless, missing paths, checkpoints and the original equity-price panel prevent complete end-to-end reproduction.

Keywords: ARCH(∞); volatility kernels; weak identification; QLIKE; predictive calibration; GJR-GARCH; stochastic volatility.
JEL: C22, C52, G17.

1 Introduction and contributions

Volatility clustering motivates the ARCH and GARCH classes introduced by Engle (1982) and Bollerslev (1986). Persistent volatility can also be represented by heterogeneous or fractional-lag constructions (Baillie et al. 1996; Müller et al. 1997; Corsi 2009). DARM writes ARCH feedback as an aggregate coefficient times a normalised lag distribution. This improves interpretability and makes different lag-decay geometries easy to compare; it does not create a class outside constrained ARCH. Existing ARCH(∞) stationarity and dependence theory already provides much of the underlying mathematics (Giraitis et al. 2000; Davidson 2004; Robinson and Zaffaroni 2006).

The main contribution is to connect the *geometry* of a normalised finite power-law kernel with a precise notion of response persistence and with finite-sample weak identification, then test whether those properties have empirical relevance. We develop (i) explicit causal and higher-moment conditions

and the exact geometric GARCH/GJR correspondence, (ii) finite-lag renewal-pole and ideal-weight identification results, alongside a kernel-score metric that explains a near-flat parameter direction, and (iii) distinct empirical evaluations of point-forecast loss and calibration of predictive distributions for observable versus latent variance. Longer proofs, exploratory comparisons and experiment-specific tables are placed in an accompanying technical supplement. No trading-profitability claim is made.

2 Model specification and assumptions

Let $(z_t)_{t\in\mathbb{Z}}$ be i.i.d., defined on a probability space, and let $\mathcal{F}_t = \sigma(z_s : s \le t)$. Assume $\mathbb{E}z_0 = 0$ and $\mathbb{E}z_0^2 = 1$. A causal model has $h_t > 0$ measurable with respect to $\mathcal{F}_{t-1}$ and

$$\varepsilon_t = \sqrt{h_t}\, z_t, \qquad \mathbb{E}(\varepsilon_t \mid \mathcal{F}_{t-1}) = 0, \qquad \mathbb{E}(\varepsilon_t^2 \mid \mathcal{F}_{t-1}) = h_t. \tag{1}$$

For finite $K \ge 1$, let $w_k(\theta) \ge 0$ and $\sum_{k=1}^{K} w_k(\theta) = 1$. The symmetric DARM model is

$$h_t = \omega + \rho \sum_{k=1}^{K} w_k(\theta)\varepsilon_{t-k}^2, \qquad \omega > 0, \quad 0 \le \rho < 1. \tag{2}$$

The infinite version replaces K by ∞ and requires absolute normalisation of the nonnegative weights; it is a separate process specification. In the finite case the conventional ARCH coefficients are $a_k = \rho w_k$. Conversely, any nonnegative finite ARCH coefficient vector with $\sum_k a_k < 1$ is recoverable as $\rho = \sum_k a_k$ and $w_k = a_k/\rho$ for $\rho > 0$. Thus the unrestrained finite class is exactly ARCH(K); DARM imposes low-dimensional restrictions on the kernel.

With nonnegative kernels $h_t \ge \omega$ whenever the series defining it exists. The total feedback ρ and timing distribution w are distinct parameters *in the mathematical parameterisation*, but finite samples can still confound them statistically, particularly at weak feedback.

2.1 Asymmetric extension

Define

$$a_\kappa(z) = z^2\big(1 - \kappa + 2\kappa \mathbf{1}_{\{z<0\}}\big), \qquad 0 \le \kappa < 1, \tag{3}$$

$$u_t = \varepsilon_t^2\big(1 - \kappa + 2\kappa \mathbf{1}_{\{\varepsilon_t<0\}}\big) = h_t a_\kappa(z_t), \tag{4}$$

$$h_t = \omega + \rho \sum_{k\ge 1} w_k u_{t-k}. \tag{5}$$

Here the generic upper summation limit stands for either K or ∞. Holding $|\varepsilon|$ and the other lags fixed, a negative shock contributes the factor $1 + \kappa$ and a positive shock the factor $1 - \kappa$; their ratio is $(1 + \kappa)/(1 - \kappa)$. It diverges near $\kappa = 1$ and is therefore a fragile point estimate at the boundary. This is *news-impact asymmetry*; it need not establish an economic leverage channel or causal stock-return/volatility relationship.

3 Causal stationarity and moments

Write $A_t = a_\kappa(z_t)$ in the asymmetric case and $A_t = z_t^2$ in the symmetric case. Let $m = \mathbb{E}A_t \in (0, \infty)$. Then $u_t = A_t h_t$ and $h_t = \omega + \rho \sum_k w_k u_{t-k}$. The common formulation unifies the proofs.

Theorem 3.1 (Causal stationary solution with finite mean)**.** *Suppose A_t is a nonnegative measurable function of z_t with $0 < m = \mathbb{E}A_t < \infty$, the weights satisfy $w_k \ge 0$ and $\sum_{k\ge1} w_k = 1$, and $\rho m < 1$.*

Then the finite- or infinite-lag equation admits a causal, almost surely finite, strictly stationary and ergodic solution with $h_t \geq \omega > 0$. It is unique among causal, strictly stationary solutions having finite $\mathbb{E}h_t$ and constructed with the same innovation sequence. Moreover,

$$\mathbb{E}h_t = \frac{\omega}{1-\rho m}, \qquad \mathbb{E}u_t = \frac{m\omega}{1-\rho m}. \tag{6}$$

Proof. Set $u_t^{(0)} = \omega A_t$ and recursively define the nonnegative path expansion

$$u_t = \omega \sum_{n=0}^{\infty} \rho^n \sum_{k_1,\dots,k_n \geq 1} \left(\prod_{i=1}^{n} w_{k_i} \right) \prod_{j=0}^{n} A_{t-\sum_{i=1}^{j} k_i}, \tag{7}$$

where the $n = 0$ term equals ωA_t. For each multi-index, the innovation dates in the product are distinct, so independence gives the expected n-th term $\omega m(\rho m)^n$. By Tonelli, $\mathbb{E}u_t = \omega m/(1-\rho m) < \infty$. Therefore the nonnegative series converges almost surely, and substituting it into the recursion verifies the model. Define $h_t = \omega + \rho \sum_k w_k u_{t-k}$; this is finite almost surely, predictable, positive and has mean in (6). The solution is a measurable shift-equivariant function of an i.i.d. two-sided sequence, hence strictly stationary and ergodic.

For another finite-mean causal stationary solution driven by the same A_t, write the difference at time t as $|u_t - \tilde{u}_t| \leq \rho A_t \sum_k w_k |u_{t-k} - \tilde{u}_{t-k}|$. The lagged terms are independent of A_t, and stationarity implies $\mathbb{E}|u_t - \tilde{u}_t| \leq \rho m \mathbb{E}|u_t - \tilde{u}_t|$. Since $\rho m < 1$, the expectation vanishes; the two solutions coincide almost surely. This proves uniqueness in the specified class, not a blanket uniqueness claim for arbitrary nonintegrable solutions. □

Corollary 3.2 (Symmetric innovations and unconditional variance)**.** *If $\kappa = 0$, $m = \mathbb{E}z_t^2 = 1$. More generally if the innovation law is symmetric about zero, then $\mathbb{E}[z_t^2 \mathbf{1}_{\{z_t<0\}}] = 1/2$ and $m = 1$ for all $0 \leq \kappa < 1$. Consequently $\mathbb{E}h_t = \mathbb{E}\varepsilon_t^2 = \omega/(1-\rho)$, and returns are uncorrelated with constant finite variance.*

Proof. Directly,

$$m = 1 + \kappa\big(2\mathbb{E}[z_t^2 \mathbf{1}_{\{z_t<0\}}] - 1\big).$$

Symmetry gives $m = 1$. Independence of z_t and the predictable h_t implies $\mathbb{E}\varepsilon_t^2 = \mathbb{E}h_t$ and $\mathbb{E}(\varepsilon_t \mid \mathcal{F}_{t-1}) = 0$, so $\mathbb{E}\varepsilon_t\varepsilon_{t-j} = 0$ for every $j > 0$ by iterated conditioning and Cauchy–Schwarz. □

Remark 3.3 (Scope of the conditions)**.** The inequality $\rho m < 1$ is a sufficient condition for an integrable causal stationary solution and yields the stated mean. Under stationarity and integrability with $\omega > 0$ it is also necessary, by taking expectations in the recursion. It is *not* a general necessary condition for strict stationarity without finite moments; familiar GARCH counterexamples are discussed by Nelson (1990). The imposed $\rho < 1$ alone does not guarantee the asymmetric variance formula under skewed innovations unless $m \leq 1/\rho$ is also ensured. Independence, not merely zero unconditional correlation between innovations, is used in these proofs.

3.1 Fourth and higher moments

Proposition 3.4 (Simple sufficient fourth-moment criterion)**.** *Suppose $\mathbb{E}z_0^4 < \infty$ and $\rho(\mathbb{E}A_0^2)^{1/2} < 1$. Then $\mathbb{E}h_t^2$ and $\mathbb{E}\varepsilon_t^4$ are finite. More generally, if $q \geq 1$, $\mathbb{E}A_0^q < \infty$ and $\rho(\mathbb{E}A_0^q)^{1/q} < 1$, then $\mathbb{E}h_t^q < \infty$.*

Proof. Apply the L^q norm to finite truncations of the nonnegative series (7). Each product of $n+1$ distinct independent factors has L^q norm $(\mathbb{E}A_0^q)^{(n+1)/q}$. Minkowski's inequality, nonnegative weights summing to one, and the triangle inequality bound the norm by the convergent geometric series $\omega\|A_0\|_q/(1-\rho\|A_0\|_q)$. Monotone convergence establishes $u_t \in L^q$ and hence $h_t \in L^q$. For $q = 2$, z_t is independent of h_t, so $\mathbb{E}\varepsilon_t^4 = \mathbb{E}h_t^2 \mathbb{E}z_t^4 < \infty$. □

This norm bound is deliberately conservative. At $K = 1$, $h_t = \omega + \rho A_{t-1} h_{t-1}$ gives, with $\omega > 0$ and $\mathbb{E}A^2 < \infty$, the exact finite-second-moment threshold $\rho^2 \mathbb{E}A^2 < 1$; the first moment already follows because $\mathbb{E}A \le (\mathbb{E}A^2)^{1/2}$. For larger finite K, stronger sufficient criteria can be obtained from the spectral radius of the second-moment companion matrix described in the accompanying technical supplement. Heavy-tailed Student-t innovations with degrees of freedom $2 < \nu \le 4$ have finite conditional variance but infinite innovation fourth moment, so fourth-moment and covariance claims cannot then be invoked automatically.

4 Relation to established GARCH models

Theorem 4.1 (Exact geometric DARM/GARCH equivalence)**.** *Let $w_k = (1-\beta)\beta^{k-1}$, $0 \le \beta < 1$, on* infinitely many *lags. Symmetric DARM is algebraically equivalent, for its integrable causal stationary solution, to*

$$h_t = \omega_G + \alpha_G \varepsilon_{t-1}^2 + \beta h_{t-1}, \quad \omega_G = (1-\beta)\omega_D, \quad \alpha_G = \rho(1-\beta). \tag{8}$$

Conversely, every GARCH(1,1) with $\omega_G > 0$, $\alpha_G \ge 0$, $\beta \in [0,1)$ and $\alpha_G + \beta < 1$ has the finite-variance geometric-DARM representation

$$\rho = \frac{\alpha_G}{1-\beta}, \qquad \omega_D = \frac{\omega_G}{1-\beta}.$$

Proof. Write $h_t = \omega_D + \rho(1-\beta)\sum_{k\ge1}\beta^{k-1}\varepsilon_{t-k}^2$. Subtract βh_{t-1}; the lagged weighted sums cancel, leaving $(1-\beta)\omega_D + \rho(1-\beta)\varepsilon_{t-1}^2$. For the reverse direction, iterate (8) n times to obtain

$$h_t = \omega_G \sum_{j=0}^{n-1} \beta^j + \alpha_G \sum_{k=1}^{n} \beta^{k-1}\varepsilon_{t-k}^2 + \beta^n h_{t-n}.$$

The remainder converges to zero in L^1 under the finite-mean stationary assumptions, because $\mathbb{E}|\beta^n h_{t-n}| = \beta^n \mathbb{E}h_0 \to 0$. This yields the infinite kernel and parameter mapping. The restrictions give $0 \le \rho < 1$ and the variance identity $\omega_D/(1-\rho) = \omega_G/(1-\alpha_G-\beta)$. □

Truncation at finite K destroys the exact GARCH recursion: the shifted geometric sum leaves an extra term proportional to $\beta^K \varepsilon_{t-K-1}^2$. Re-normalisation changes all coefficients, not merely the omitted tail. This distinction corrects the interpretation of the existing preliminary derivation: its intercept and parameter mapping are valid for the infinite kernel, while the implemented $K = 250$ process is not exactly GARCH.

Proposition 4.2 (Asymmetric geometric kernel and restricted GJR-GARCH)**.** *With symmetric innovations and an infinite geometric kernel, asymmetric DARM satisfies*

$$h_t = \omega_G + \big(\alpha_G + \gamma_G \mathbf{1}_{\{\varepsilon_{t-1}<0\}}\big)\varepsilon_{t-1}^2 + \beta h_{t-1} \tag{9}$$

for $\alpha_G = \rho(1-\beta)(1-\kappa)$, $\gamma_G = 2\rho(1-\beta)\kappa$, and $\omega_G = (1-\beta)\omega_D$. Its expected persistence is $\alpha_G + \gamma_G/2 + \beta = \rho(1-\beta) + \beta < 1$ for symmetric innovations.

Proof. The geometric sum telescopes exactly as in Theorem 4.1, with the squared shock ε_t^2 replaced by the transformed shock u_t. Expanding the negative-shock multiplier yields the two displayed coefficients and the persistence identity. □

For positive feedback, the inverse mapping is $\rho = (\alpha_G + \gamma_G/2)/(1-\beta)$ and $\kappa = \gamma_G/(2\alpha_G + \gamma_G)$, admissible precisely when $\alpha_G > 0$, $\gamma_G \geq 0$, $\beta \in [0,1)$ and $\alpha_G + \gamma_G/2 + \beta < 1$; it also permits the zero-feedback case $\alpha_G = \gamma_G = 0$, with $\rho = 0$ and κ unidentified. The case $\alpha_G = 0, \gamma_G > 0$ requires $\kappa = 1$ and is therefore excluded. Negative-γ_G GJR models and cases with $\alpha_G < 0$ are likewise outside this DARM domain. Thus geometric asymmetric DARM is not in itself a structural extension of this restricted benchmark (Glosten et al. 1993).

5 Kernels and the meaning of memory

For $k = 1, \ldots, K$, let

$$w_k^{\exp}(\beta) = \frac{\beta^{k-1}}{\sum_{j=1}^{K} \beta^{j-1}}, \qquad 0 < \beta < 1, \tag{10}$$

$$w_k^{\mathrm{pow}}(\alpha, \delta) = \frac{(k-1+\delta)^{-\alpha}}{\sum_{j=1}^{K}(j-1+\delta)^{-\alpha}}, \qquad \alpha, \delta > 0, \tag{11}$$

$$w_k^{\mathrm{hyp}}(s, p) = \frac{(1+sk)^{-p}}{\sum_{j=1}^{K}(1+sj)^{-p}}, \qquad s, p > 0. \tag{12}$$

All are finite positive normalised weights. The hyperbolic form used in the exploratory notebook is exactly a shifted power-law kernel with $\alpha = p$ and $\delta = 1 + 1/s$, since the constant factor s^{-p} cancels under normalisation. It therefore covers a subset $\delta > 1$ of the power-law parameter space, not an independent decay family. Their nearly identical validation performance is consistent with this nesting.

For $K = \infty$, exponential weights have the geometric normalisation $(1-\beta)\beta^{k-1}$. The infinite power-law (and hyperbolic) normalisation exists *if and only if* $\alpha > 1$ (and $p > 1$, respectively), by comparison with the Riemann p-series. The finite-lag specification is well defined even when $\alpha \leq 1$, but no infinite probability kernel exists with the same raw weights in that case.

A simple effective-memory summary is the Herfindahl inverse $K_{\mathrm{eff}} = (\sum_k w_k^2)^{-1} \in [1, K]$; a separate average lag $\bar{k} = \sum_k k w_k$ may be finite or infinite for an infinite power-law kernel, depending on whether $\alpha > 2$. These are descriptive summaries, not formal dependence orders. Parameters can be weakly identifiable when $\rho \approx 0$, $K = 1$, $\alpha \approx 0$, or δ is very large relative to K.

5.1 Summability of squared-return dependence

Let $m = \mathbb{E}A_0$ and $c_k = \rho m w_k$; assume $\mathbb{E}u_t^2 < \infty$, $\rho m < 1$ and the conditions of Theorem 3.1. Set $\zeta_t = u_t - m h_t$ (a square-integrable martingale difference) and $Y_t = u_t - \mathbb{E}u_t$. Then

$$Y_t = \zeta_t + \sum_{k \geq 1} c_k Y_{t-k}. \tag{13}$$

Define $\psi_0 = 1$ and $\psi_j = \sum_{k=1}^{j} c_k \psi_{j-k}$ for $j \geq 1$. The convolution-renewal sequence is nonnegative and $\sum_{j \geq 0} \psi_j = (1-\rho m)^{-1}$, because it is the convolution series $\sum_{n \geq 0} c^{*n}$.

Theorem 5.1 (No non-summable covariance under finite fourth moments)**.** *Under the preceding conditions the unique integrable solution has the L^2 representation*

$$Y_t = \sum_{j \geq 0} \psi_j \zeta_{t-j}, \qquad \sum_{\ell \in \mathbb{Z}} |\operatorname{Cov}(Y_0, Y_\ell)| \leq \operatorname{Var}(\zeta_0) \Big(\sum_{j \geq 0} \psi_j \Big)^2 < \infty. \tag{14}$$

The covariance function of the centred squared returns $\varepsilon_t^2 - \mathbb{E}\varepsilon_t^2$ is also absolutely summable if $\mathbb{E}\varepsilon_t^4 < \infty$. For finite K, the coefficients ψ_j decay at least geometrically. Thus neither finite DARM nor an integrable

infinite power-law DARM with finite fourth moments has genuine long memory in the sense of non-summable squared-return autocovariances.

Proof. The innovation ζ_t has conditional mean zero and is orthogonal across distinct dates. Since $\sum c_k < 1$, the sequence ψ is in $\ell^1 \subset \ell^2$, and the series in (14) converges in L^2 and solves (13). The difference between this solution and the stationary L^2 solution has zero L^2 norm by a contraction inequality: $\|D_t\|_2 \le (\sum c_k)\|D_0\|_2$. Orthogonality now gives $\mathrm{Cov}(Y_t, Y_{t+\ell}) = \mathrm{Var}(\zeta_0)\sum_j \psi_j\psi_{j+|\ell|}$; absolute summability follows by Tonelli.

Put $\chi_t = \varepsilon_t^2 - h_t$, another martingale difference, square integrable under the fourth-moment assumption. The crucial identity is obtained *before* removing the lag-zero term. Since $Y_t = \zeta_t + m(h_t - \mathbb{E}h_t)$ and $\psi_0 = 1$,

$$h_t - \mathbb{E}h_t = \frac{Y_t - \zeta_t}{m} = \frac{1}{m}\left(\sum_{j\ge 0}\psi_j\zeta_{t-j} - \zeta_t\right) = \frac{1}{m}\sum_{j\ge 1}\psi_j\zeta_{t-j}. \tag{15}$$

There is *no* additional subtraction of ζ_t/m once the sum starts at $j = 1$. Consequently,

$$\varepsilon_t^2 - \mathbb{E}\varepsilon_t^2 = \chi_t + \sum_{j\ge 1}\frac{\psi_j}{m}\zeta_{t-j}. \tag{16}$$

Equivalently, the conditional-variance filter is $D_m(s) = \rho W(s)/[1 - \rho m W(s)] = [\Psi_m(s) - 1]/m$ where $W(s) = \sum_{k\ge 1} w_k s^k$ and $\Psi_m(s) = [1 - \rho m W(s)]^{-1}$. Its lag-zero coefficient is zero, and its lag-j coefficient is ψ_j/m for $j \ge 1$. The coefficients are absolutely summable. The contemporaneous pair (χ_t, ζ_t) may be correlated but is orthogonal to pairs at other dates. Cauchy–Schwarz and double summation show that the squared-return autocovariances are absolutely summable. Finally, when $K < \infty$, choose $R > 1$ sufficiently close to 1 so that $\sum_{k=1}^K c_k R^k < 1$. The power-series coefficients obey $\sum_j \psi_j R^j < \infty$, hence $\psi_j = O(R^{-j})$. □

Remark 5.2. The theorem does *not* exclude slow, practically persistent hyperbolic-looking autocorrelations over moderate samples; nor does it address nonintegrable boundary regimes, infinite-fourth-moment covariance, or nonlinear notions of long memory. Indeed, results for general ARCH(∞) models show power-decaying dependence that can mimic long memory (Giraitis et al. 2000; Davidson 2004). For $K < \infty$ the covariance tail is ultimately geometric irrespective of the lag-weight shape.

6 Kernel geometry, exact persistence and identifiability

The normalised representation admits explicit mathematical results about the relation between kernel shape, forecast persistence and higher moments. The results in this section are derived for DARM's parameterisation, but their underlying renewal and random-recursion principles are not presented as new general ARCH theory. The accompanying technical supplement supplies further derivations and Monte Carlo tables.

6.1 Sensitivity of the power-law memory distribution

For finite K, put $x_k = k - 1 + \delta$, $w_k = x_k^{-\alpha}/Z$, and write $\langle g\rangle_w = \sum_k w_k g(k)$. Elementary differentiation gives

$$\frac{\partial w_k}{\partial \alpha} = w_k(\langle \log x\rangle_w - \log x_k), \qquad \frac{\partial w_k}{\partial \delta} = -\alpha w_k(x_k^{-1} - \langle x^{-1}\rangle_w). \tag{17}$$

Proposition 6.1 (Monotonicity of memory length and concentration). *For $K \geq 2$ and $\alpha, \delta > 0$, the mean lag $\bar{k} = \sum_k w_k k$ strictly decreases with α and strictly increases with δ. The effective count $K_{\text{eff}} = (\sum_k w_k^2)^{-1}$ has the same directional dependence.*

Proof. From (17), $\partial_\alpha \bar{k} = -\operatorname{Cov}_w(k, \log x_k) < 0$ and $\partial_\delta \bar{k} = -\alpha \operatorname{Cov}_w(k, x_k^{-1}) > 0$, since k and $\log x_k$ are strictly increasing and x_k^{-1} strictly decreasing. Define $q_k = w_k^2 / \sum_j w_j^2$. Then $\partial_\alpha \log(\sum_k w_k^2) = 2[\langle \log x \rangle_w - \langle \log x \rangle_q] > 0$ and $\partial_\delta \log(\sum_k w_k^2) = 2\alpha[\langle x^{-1} \rangle_w - \langle x^{-1} \rangle_q] < 0$, because $q_k / w_k \propto w_k$ decreases with k, making q more concentrated at earlier lags. Taking the reciprocal concludes the proof. □

K_{eff} quantifies the concentration of the normalised kernel, but does not itself measure a temporal half-life. For $K = \infty$ the normalisation is finite iff $\alpha > 1$, while $\bar{k}$ is finite only if $\alpha > 2$.

6.2 Renewal response and dominant finite-lag pole

Define $C(s) = \rho \sum_{k=1}^{K} w_k s^k$ and $\Psi(s) = 1/[1 - C(s)] = \sum_{j \geq 0} \psi_j s^j$. The renewal recursion $\psi_0 = 1$, $\psi_j = \rho \sum_{k=1}^{\min(K,j)} w_k \psi_{j-k}$ implies $\psi_j \geq 0$ and $\sum_j \psi_j = (1-\rho)^{-1}$. A useful interpretation is that $(1-\rho)\psi_j$ gives the distribution of a sum of an independent geometric number of random lags drawn from w; if $\bar{k} < \infty$ the mean total propagation delay is $\rho\bar{k}/(1-\rho)$.

Theorem 6.2 (Exact finite-K asymptotic response). *Suppose $K < \infty$, $0 < \rho < 1$, $w_1 > 0$ and $w_k \geq 0$ with $\sum_k w_k = 1$. The equation $C(R) = 1$ has a unique solution $R > 1$, and $s = R$ is the only zero of $1 - C(s)$ in the closed disk $|s| \leq R$. Moreover,*

$$\psi_j \sim \frac{R^{-j}}{RC'(R)}, \qquad \rho^{-1/K} \leq R \leq \rho^{-1/\bar{k}} \leq \rho^{-1}. \tag{18}$$

For shifted power-law weights R increases strictly with α and decreases strictly with δ, ρ and K (the last at fixed kernel-shape parameters). At fixed K, $\log R \sim (1-\rho)/\bar{k}$ as $\rho \uparrow 1$.

Proof. $C(r)$ is increasing for $r > 0$, $C(1) = \rho < 1$ and $C(r) \to \infty$, giving the unique positive R. For $|s| < R$, $|C(s)| \leq C(|s|) < 1$; on $|s| = R$, equality and $C(s) = 1$ force $s = R$ because $w_1 > 0$. This pole is simple as $C'(R) > 0$. Standard Taylor-coefficient extraction of its principal part proves the asymptotic equivalent. The bounds follow from $R \leq \mathbb{E}_w R^k \leq R^K$ and Jensen's inequality $\mathbb{E}_w R^k \geq R^{\bar{k}}$. Implicit differentiation gives $\partial_\alpha R = \operatorname{Cov}_w(R^k, \log x_k)/W'(R) > 0$ and $\partial_\delta R = \alpha \operatorname{Cov}_w(R^k, x_k^{-1})/W'(R) < 0$. Increasing ρ or appending a strictly later lag increases $C(R)$ and so decreases its root. Taylor expansion at $R = 1$ gives the boundary approximation. □

This pole gives a *long-horizon* response half-life $\log 2 / \log R$, not necessarily the exact decay rate over short horizons. Figure 1 shows examples at common ρ but different kernel shapes. When $K = \infty$ and $w_j \sim c j^{-\alpha}$ with $\alpha > 1$, the classical local-subexponential defective renewal theorem instead gives $\psi_j \sim \rho(1-\rho)^{-2} w_j$ (Foss et al. 2013). This is a *summable* power-law tail, not proof of non-summable long memory. The order of $K \to \infty$ and $j \to \infty$ matters.

6.3 Exact identification of the kernel-shape mapping

Theorem 6.3 (Ideal-weight identification). *For finite $K \geq 3$ and $\alpha, \delta > 0$, the vector $(w_1, \ldots, w_K)$ uniquely determines (α, δ); its Jacobian has rank two. This fails at $\alpha = 0$ and for $K \leq 2$.*

Proof. Let $f(x) = \log(1 + 1/x)$. Then $r_1 = \log(w_1/w_2) = \alpha f(\delta)$ and $r_2 = \log(w_2/w_3) = \alpha f(\delta+1)$. The function f is the Laplace transform of $(1 - e^{-t})/t \, dt$ on $(0, \infty)$, so $\log f$ is strictly convex (its

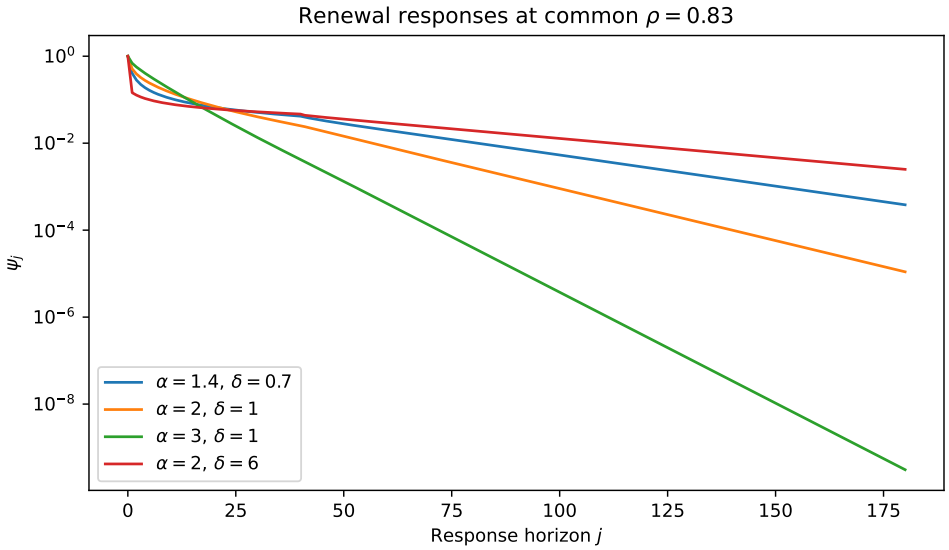


Figure 1: Theoretical renewal multipliers for four power-law kernels, $K = 40$ and $\rho = 0.83$, calculated from the exact coefficient recursion. They are not market-data estimates.

second derivative is the variance under exponential tilting). It follows that $f(x)/f(x+1)$ is strictly decreasing, from infinity to one. Therefore r_1/r_2 uniquely identifies δ, and then $r_1/f(\delta)$ identifies α. The determinant of the (r_1, r_2) Jacobian with respect to (α, δ) is $\alpha[f(\delta)f'(\delta+1) - f(\delta+1)f'(\delta)] > 0$. One or two weights provide at most one independent ratio; at $\alpha = 0$ all weights are equal regardless of δ. □

This concerns a *known coefficient vector*, not recovery from the law of ε_t with unknown z_t and a finite sample. Weak identification may arise for small ρ, small α, or $\delta \gg K$; expanding $\log(k - 1 + \delta)$ then shows that many (α, δ) pairs with similar α/δ induce similar weights. Formal likelihood identification and asymptotic estimation require additional hypotheses (Robinson and Zaffaroni 2006).

7 Fourth-moment and spectral refinements

For a symmetric innovation distribution let $\mu_4 = \mathbb{E}z_t^4$ and retain the asymmetric shock transform $A_t = z_t^2[1 - \kappa + 2\kappa \mathbf{1}_{\{z_t<0\}}]$ from Section 2. Provided $\mathbb{E}h_t^2 < \infty$, the transformed and actual squared-return innovations

$$\zeta_t = u_t - h_t = h_t(A_t - 1), \qquad \chi_t = \varepsilon_t^2 - h_t = h_t(z_t^2 - 1)$$

are martingale differences. Both are orthogonal across dates but have a generally nonzero *contemporaneous* cross-covariance.

Proposition 7.1 (Asymmetry and noise fourth moments)**.** *Under symmetric standardised innovations,* $\mathbb{E}A_t = 1$ *and*

$$\begin{aligned} \mathbb{E}A_t^2 &= \mu_4(1+\kappa^2), & \operatorname{Var}\zeta_t &= [\mu_4(1+\kappa^2) - 1]\mathbb{E}h_t^2, \\ \operatorname{Var}\chi_t &= (\mu_4 - 1)\mathbb{E}h_t^2, & \operatorname{Cov}(\zeta_t, \chi_t) &= (\mu_4 - 1)\mathbb{E}h_t^2. \end{aligned} \tag{19}$$

Proof. Sign symmetry implies $\mathbb{E}[z_t^{2p}\mathbf{1}_{\{z_t<0\}}] = \frac{1}{2}\mathbb{E}z_t^{2p}$ whenever the moment exists. This yields $\mathbb{E}A^2 = \mu_4[(1-\kappa)^2 + (1+\kappa)^2]/2$ and $\mathbb{E}[(A-1)(z^2-1)] = \mu_4 - 1$. Independence of h_t and current z_t gives (19). □

Under the symmetric innovation law $m = 1$, let $C(s) = \rho \sum_k w_k s^k$ and $D(s) = D_1(s) = C(s)/[1 - C(s)]$ as obtained in (15). The moving-average relations are

$$u_t - \mathbb{E}u_t = [1 - C(L)]^{-1}\zeta_t, \qquad h_t - \mathbb{E}h_t = D(L)\zeta_t, \qquad \varepsilon_t^2 - \mathbb{E}\varepsilon_t^2 = \chi_t + D(L)\zeta_t. \tag{20}$$

Consequently the spectra under square integrability are

$$f_h(\lambda) = \frac{\operatorname{Var}\zeta_t}{2\pi}|D(e^{-i\lambda})|^2, \tag{21}$$

$$f_{\varepsilon^2}(\lambda) = \frac{1}{2\pi}\left\{\operatorname{Var}\chi_t + \operatorname{Var}\zeta_t|D(e^{-i\lambda})|^2 + 2\operatorname{Cov}(\zeta_t, \chi_t)\Re D(e^{-i\lambda})\right\}. \tag{22}$$

The cross term is necessary: modelling the transformed signal u_t and the actual squared returns ε_t^2 as having identical spectra would be incorrect for nonzero κ. The stationary transformed-squared-shock autocovariance satisfies $\gamma_u(\ell) = \rho \sum_k w_k \gamma_u(\ell - k)$ for positive integer ℓ (with even extension to negative lags). Both covariance sequences are absolutely summable under the displayed moment assumptions, consistent with established ARCH(∞) results (Giraitis et al. 2000; Davidson 2004).

For finite K, the transformed state $\boldsymbol{S}_t = (u_t, \ldots, u_{t-K+1})^\top$ obeys the random affine recursion $\boldsymbol{S}_t = \boldsymbol{M}_t\boldsymbol{S}_{t-1} + \omega A_t \boldsymbol{e}_1$, where the first row of $\boldsymbol{M}_t$ is $A_t\rho(w_1, \ldots, w_K)$ and its subdiagonal is one. Under $\mathbb{E}\log^+\|\boldsymbol{M}_0\| < \infty$, $\mathbb{E}\log^+(\omega A_0) < \infty$ and a negative top Lyapunov exponent for products of $(\boldsymbol{M}_t)$, the backward random-affine construction yields a causal strictly stationary solution (assuming the required nondegeneracy for uniqueness) (Bougerol and Picard 1992). A separate, stronger and computationally useful sufficient condition for finite fourth return moments is

$$\operatorname{spr}(\mathbb{E}[\boldsymbol{M}_t \otimes \boldsymbol{M}_t]) < 1, \qquad \mathbb{E}z_t^4 < \infty, \tag{23}$$

when $\rho\mathbb{E}A < 1$. It follows by vectorising the affine state second-moment recursion, retaining the within-time dependence of $\boldsymbol{M}_t$ and $\omega A_t \boldsymbol{e}_1$. Necessity in unrestricted finite K is *not* claimed without nondegeneracy conditions. For $K = 1$, $\rho^2\mathbb{E}A^2 < 1$ is exactly the finite-$\mathbb{E}h_t^2$ criterion for the causal positive-intercept solution, and

$$\mathbb{E}h_t^2 = \frac{\omega^2(1 + \rho m)}{(1 - \rho m)(1 - \rho^2\mathbb{E}A^2)}, \qquad m = \mathbb{E}A.$$

For standardised Student-t_ν errors the fourth moment $\mu_4 = 3(\nu - 2)/(\nu - 4)$ exists only when $\nu > 4$; $\nu > 2$ suffices for variance but not covariance-spectral assertions about squared returns. See the accompanying technical supplement for additional Kronecker-matrix details and diagnostics.

8 Finite-state multi-step forecast representation

The recursion in Section 9 admits a compact companion-matrix solution. Let $\boldsymbol{a} = \rho(w_1, \ldots, w_K)^\top$, $\boldsymbol{s}_t = (u_t, \ldots, u_{t-K+1})^\top$, and $\boldsymbol{B}$ be the deterministic companion with first row $m\boldsymbol{a}^\top$ and unit subdiagonal, where $m = \mathbb{E}A_t$. Let $\boldsymbol{d} = m\omega\boldsymbol{e}_1$. For $j \geq 1$,

$$f_j = \mathbb{E}_t h_{t+j} = \omega + \boldsymbol{a}^\top\left(\boldsymbol{B}^{j-1}\boldsymbol{s}_t + \sum_{r=0}^{j-2}\boldsymbol{B}^r\boldsymbol{d}\right), \tag{24}$$

$$f_j - \frac{\omega}{1 - \rho m} = \boldsymbol{a}^\top\boldsymbol{B}^{j-1}\left(\boldsymbol{s}_t - \frac{m\omega}{1 - \rho m}\mathbf{1}\right). \tag{25}$$

This follows by iterating $\mathbb{E}_t\boldsymbol{s}_{t+j} = \boldsymbol{B}\mathbb{E}_t\boldsymbol{s}_{t+j-1} + \boldsymbol{d}$, using independence of A_{t+j} from the future predictable variance. When $\rho m < 1$ and all weights are positive, the Perron eigenvalue is $r_B = 1/R_m < 1$ where R_m solves $\rho m \sum_k w_k R_m^k = 1$; thus the conditional variance mean-reverts geometrically for fixed finite K, with an eventual rate derived from its kernel. At symmetric innovations, $m = 1$ even if $\kappa > 0$, but signed realised past shocks enter through the initial vector $\boldsymbol{s}_t$. GARCH's single scalar forecast mode is recovered under an infinite geometric kernel; general finite DARM has multiple companion modes.

9 Forecast recursion and estimation

Let $\mathbb{E}_t(\cdot) = \mathbb{E}(\cdot \mid \mathcal{F}_t)$, with h_{t+1} known at the end of day t. Write $f_j = \mathbb{E}_t h_{t+j}$, $j \geq 1$, and $v_s = u_s$ for $s \leq t$. For $j \geq 1$,

$$f_j = \omega + \rho \left[\sum_{k \geq j} w_k u_{t+j-k} + m \sum_{k=1}^{j-1} w_k f_{j-k} \right], \qquad V_{t,H} = \sum_{j=1}^{H} f_j. \tag{26}$$

The finite case uses $k \leq K$ and sets an empty sum to zero. To derive the recursion, split past observed inputs ($t + j - k \leq t$) from future inputs. Independence of A_{t+j-k} from h_{t+j-k} gives $\mathbb{E}_t u_{t+j-k} = m \mathbb{E}_t h_{t+j-k}$ for future dates. Under symmetric innovations $m = 1$, for both symmetric and sign-asymmetric DARM. Formula (26) requires a finite mean and valid causal conditioning. If future returns satisfy $\mathbb{E}(\varepsilon_{t+j} \mid \mathcal{F}_{t+j-1}) = 0$, then their cross-products have zero conditional expectation, and

$$\mathrm{Var}_t\Big(\sum_{j=1}^{H} \varepsilon_{t+j} \Big) = \sum_{j=1}^{H} \mathbb{E}_t h_{t+j} = V_{t,H}. \tag{27}$$

This equality is not automatically a forecast of portfolio variance when cross-asset covariance and dynamic portfolio weights are present.

9.1 Likelihood and implementation

For the reported experiments $K = 250$ and normalised shifted power-law weights, with standardised Student-t innovations. On percentage-scaled returns $x_t = 100\varepsilon_t$, conditional variance $H_t = 10^4 h_t$, the log likelihood contribution (omitting no parameter-dependent terms) is

$$\begin{aligned} \ell_t = & -\tfrac{1}{2} \log H_t + \log \Gamma\left(\frac{\nu+1}{2} \right) - \log \Gamma\left(\frac{\nu}{2} \right) - \tfrac{1}{2} \log[(\nu-2)\pi] \\ & - \frac{\nu+1}{2} \log\left(1 + \frac{x_t^2}{(\nu-2)H_t} \right), \qquad \nu > 2. \end{aligned} \tag{28}$$

The intercept is optimised relative to the sample variance, with L-BFGS-B, parameter bounds $\rho \in [10^{-5}, 0.999]$, $\kappa \in [0, 0.999]$, $\alpha \in [10^{-4}, 10]$, $\delta \in [10^{-4}, 100]$ and $\nu \in [2.05, 100]$, supplemented by multiple starting points. The nonzero lower bound on ρ and upper bounds on asymmetry and persistence are numerical restrictions, not theoretical necessities. Only observations with fully finite lag histories enter DARM's fitted conditional likelihood.

At the end of each year 2019–2024, model parameters are fitted using data available through that December cutoff and held fixed throughout the following calendar year. The 2020–2025 forecasting recursion uses subsequently observed returns only once they become historical. The GJR implementation uses the `arch` package with zero conditional mean, GJR(1,1) variance and Student-t errors. The EWMA decay is 0.94; rolling historical variance uses 252 days. The multi-day forecasts for the two simple benchmarks use a flat daily path equal to their next-day estimate. Such a flat path is a benchmarking convention, not the exact dynamic multi-day forecast from a future-refitted rolling window.

10 Variance targets and predictive uncertainty

Let $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot \mid \mathcal{F}_t]$, where $\mathcal{F}_t = \sigma(z_s : s \leq t)$ is the innovation filtration (or any filtration relative to which the following assumptions hold). Assume h_s is finite, nonnegative and $\mathcal{F}_{s-1}$-measurable

and that

$$\mathbb{E}[z_s \mid \mathcal{F}_{s-1}] = 0, \qquad \mathbb{E}[z_s^2 \mid \mathcal{F}_{s-1}] = 1, \qquad \mathbb{E}\Big[\sum_{j=1}^{H} h_{t+j}\Big] < \infty.$$

The first condition gives martingale-difference returns, but only the *second* is needed for the equality of expected quadratic-variation targets. Define

$$RV_{t,H} = \sum_{j=1}^{H} \varepsilon_{t+j}^2, \qquad IV_{t,H} = \sum_{j=1}^{H} h_{t+j}, \qquad \varepsilon_s = \sqrt{h_s} z_s. \tag{29}$$

Proposition 10.1 (Conditional-mean agreement without predictive-law agreement)**.** *Under the preceding assumptions,*

$$\mathbb{E}_t RV_{t,H} = \mathbb{E}_t IV_{t,H}, \qquad RV_{t,H} - IV_{t,H} = \sum_{j=1}^{H} d_{t+j}, \quad d_s = h_s(z_s^2 - 1). \tag{30}$$

The sequence (d_s) is integrable and is a martingale-difference sequence with respect to $(\mathcal{F}_s)$. No independence between $RV_{t,H} - IV_{t,H}$ and the entire future variable $IV_{t,H}$ follows.

Proof. Predictability and the conditional unit second moment give $\mathbb{E}[\varepsilon_s^2 \mid \mathcal{F}_{s-1}] = h_s$ and hence $\mathbb{E}[d_s \mid \mathcal{F}_{s-1}] = 0$. Integrability follows from $\mathbb{E}|d_s| \le \mathbb{E}[h_s(z_s^2 + 1)] = 2\mathbb{E}h_s < \infty$; the required mean is finite by assumption. Summing and applying the tower property gives the equality. An $H > 1$ future value of h_{t+j} may depend on previous squared innovations $z_{t+1}^2, \dots, z_{t+j-1}^2$; hence the future IV can be statistically dependent on preceding d's. For example, in ARCH(1) with $\rho > 0$, $H = 2$, $h_{t+2} = \omega + \rho\varepsilon_{t+1}^2$, and a nondegenerate innovation fourth moment $\mu_4 = \mathbb{E}z^4 > 1$, one obtains $\mathrm{Cov}_t(RV_{t,2} - IV_{t,2}, IV_{t,2}) = \rho h_{t+1}^2(\mu_4 - 1) > 0$. Indeed h_{t+1} is $\mathcal{F}_t$-measurable and the second martingale increment is orthogonal to $IV_{t,2} \in \mathcal{F}_{t+1}$. Thus the difference need not be independent of $IV_{t,H}$. □

Under $\mathbb{E}[z_s \mid \mathcal{F}_{s-1}] = 0$ and integrability, the conditional variance of the cumulative return $\sum_{j=1}^{H} \varepsilon_{t+j}$ is $\mathbb{E}_t IV_{t,H}$. This neither identifies the *conditional distribution* of IV nor makes its predictive intervals equivalent to those for RV. For a fully observed ARCH process h_{t+1} is known at t, so the one-step IV is conditionally degenerate, although $RV_{t,1}$ is not. A latent log-SV filter may retain one-step state uncertainty. These are target-specific statements, not evidence that any fitted model attains correct calibration.

11 Empirical design and statistical procedures

The exploratory sample spans 2016–2019 after initial modelling through 2015. Results from that period motivated kernel and asymmetry choices, and must not be presented as confirmatory evidence. The reported independent period is 2020–2025, with 75 US equities, 1,508 daily test dates and 309 Friday-week-ending weekly origins for complete multi-day forecast windows. For 10-day forecasting there are $309 \times 75 = 23{,}175$ common stock-origin pairs. All five models are scored on the same finite-positive forecast mask. The original research notebook constructs daily log returns from `yfinance` prices. No independently archived raw adjusted-price panel is available in this research package; therefore all numerical results below are *audits of supplied aggregate outputs*, not reruns of the fitted models.

At origin t, the H-day realised-variance proxy is $RV_{t,H} = \sum_{j=1}^{H} \varepsilon_{t+j}^2$. The score is

$$\mathrm{QLIKE}(RV, \widehat{V}) = \log \widehat{V} + \frac{RV}{\widehat{V}}, \qquad \widehat{V} > 0. \tag{31}$$

Lower scores are preferred; negative mean QLIKE values are entirely possible because the forecast variance is measured in units of squared decimal returns. With imperfect volatility proxies, QLIKE has useful ranking robustness under conditional-unbiasedness and regularity assumptions (Patton 2011; Hansen and Lunde 2006). Nevertheless, sums of daily squared returns are not high-frequency realised variance measurements in the sense of Andersen et al. (2003), and noisy targets reduce statistical power.

For candidate C and benchmark B define daily or weekly cross-sectional loss differences $d_t = N_t^{-1} \sum_i [\mathrm{QLIKE}_{i,t}(B) - \mathrm{QLIKE}_{i,t}(C)]$. A positive $\bar{d}$ favours C. Inference is on the time-indexed d_t, rather than treating 75×309 overlapping and cross-correlated stock-origin errors as independent. Two-sided normal-approximation HAC tests use a Bartlett covariance estimator (Newey and West 1987; Diebold and Mariano 1995); the code sets ten lags for daily differences and $\max(4, \lceil H/5 \rceil + 2)$ weekly lags. The 10-day primary family comprises two candidate comparisons against GJR-GARCH, with Holm step-down multiplicity adjustment. Additional circular moving-block bootstrap confidence intervals use four-, eight- and twelve-week blocks, 2,000 draws per block length. Reported bootstrap p-values use a centred-bootstrap mean statistic, not a separate nested-model parameter test.

The supplied code visibly makes the estimation cutoff causal and checks one-step alignment between filtered and recursive DARM forecast paths. The same procedure does *not* establish formal prospective preregistration or exclude earlier research-driven choices; the statement that the primary endpoint was predeclared is taken from the supplied experiment specification. Serial dependence within overlapping windows and common market shocks across equities remain relevant even with date-level averaging.

12 Independent financial-market evidence

12.1 2020–2025 out-of-sample variance forecasts

Table 1: Independent-test mean QLIKE (2020–2025). Lower is better; one-day origins daily, other horizons weekly. The same origin–equity mask is used for all models within a horizon.

Model	1 day	5 days	10 days	20 days
Asymmetric DARM	**-7.132644**	**-5.488284**	**-4.749941**	**-3.928052**
GJR-GARCH	-7.113676	-5.467469	-4.734893	-3.923520
Symmetric DARM	-7.126926	-5.478703	-4.732278	-3.907105
EWMA	-7.080073	-5.425867	-4.677375	-3.824729
Rolling 252 days	-6.906234	-5.262699	-4.513901	-3.718481

Table 2: Independent-test QLIKE differentials against GJR-GARCH (positive favours DARM). HAC p-values are two-sided and *unadjusted* except where explicitly indicated.

Horizon	Asymmetric gain	p_{HAC}	Symmetric gain	p_{HAC}
1 day	0.018968	0.000007	0.013250	0.085023
5 days	0.020815	0.002708	0.011234	0.435530
10 days (primary)	0.015048	0.146291	-0.002615	0.912038
20 days	0.004532	0.787470	-0.016415	0.612505

Primary ten-day Holm-adjusted p: asymmetric 0.292582; symmetric 0.912038.

Asymmetric DARM has the smallest mean QLIKE at all four reported horizons (Table 1); however, the primary comparison (Table 2) is not statistically significant. The symmetric ten-day gain is slightly negative. At ten days, mean squared error (scaled by 10^8) is 6,653.061 for asymmetric DARM, 6,856.153 for GJR and 6,771.171 for symmetric DARM; the ratio of realised to predicted total variance is 1.0776

for asymmetric DARM versus 1.0963 for GJR. These descriptive quantities should not be interpreted as inferential tests.

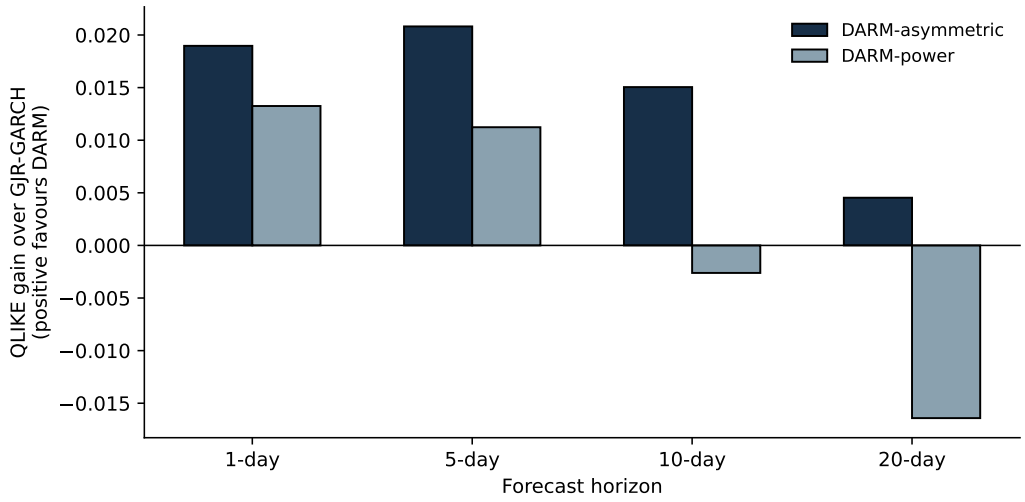


Figure 2: Independent-test mean QLIKE difference relative to Student-t GJR-GARCH. These bar heights are observed descriptive estimates; the figure does not depict confidence intervals.

The archived 4-, 8- and 12-week block-bootstrap primary confidence intervals each contain zero (the technical supplement). All primary-contrast bootstrap intervals include zero (see the accompanying technical supplement). The one- and five-day HAC findings are *secondary*; no cross-horizon multiplicity adjustment was pre-specified.

13 Synthetic identification and misspecification evidence

Experiments 13–18 use simulated DARM, GJR-GARCH and, in the later studies, latent log-SV data. Results concern the specified Monte Carlo designs only. In Experiments 13–14, 300 fits or primary profiles were reported per experiment (50 per symmetry–sample-length configuration); corresponding simulation seeds and generators overlap, so the two studies are not independent sample collections. Experiment 15 comprises 600 primary likelihood-ratio tests (100 per asymmetry–sample-length cell). Experiment 16 and Experiment 17 report 240 independent datasets *within each experiment*; Experiment 18 has 480 distinct simulated datasets, each analysed with estimated and fixed-true Student-t degrees of freedom (960 paired fitting-mode evaluations). The later selected interval/bootstrap diagnostics have much smaller effective replication counts, and their precision must not be overstated.

13.1 Weak identification and functional forecasting

The normalised power-law kernel is identified from known weights when $K \geq 3$ and (α, δ) lies in the interior (Theorem 6.3). This does not imply precise estimation from returns. Experiments 13–15 demonstrate sizeable (α, δ) recovery errors, likelihood ridges and wide δ profile sets despite improvement with training length. For example, at $T = 1500$ the symmetric Experiment 13 δ RMSE was 27.149 while fitted-versus-true variance correlation was 0.9845; at $T = 8000$ these were 2.956 and 0.9980. The primary Experiment 14 δ coverage was 94–96% in the six configurations, but symmetric $T = 1500$ median profile width was 95.156 on a parameter box capped at 100. Experiment 15 nominal-95% κ-LR coverage varied from 90% to 100% across six 100-replication cells, with frequent upper-bound estimates at true $\kappa = 0.9$. The six-cell coverage differences are exploratory; estimated

boundary hits alone do not establish that boundary-asymptotic null laws apply to interior generating parameters (Wilks 1938; Self and Liang 1987; Andrews 2001).

The log-weight derivative matrix gives a direct geometric explanation for practical ridges. With $X_k = \log(k - 1 + \delta)$, $Y_k = (k - 1 + \delta)^{-1}$ and variance/covariance under the kernel probabilities, the score Gram matrix is

$$G(\alpha, \delta) = \begin{pmatrix} \mathrm{Var}_w(X) & \alpha \,\mathrm{Cov}_w(X, Y) \\ \alpha \,\mathrm{Cov}_w(X, Y) & \alpha^2 \,\mathrm{Var}_w(Y) \end{pmatrix}. \tag{32}$$

Although G is positive definite for $K \geq 3$ at interior parameters, near-collinearity between the two score vectors can make its small eigenvalue tiny. This matrix is *not* the likelihood Fisher information for observed returns: nuisance profiling, persistence, and return-path variability affect the latter. Experiment 16 reports a theoretical response half-life of 69.78223 days at its DARM generating parameters but half-life RMSEs of approximately 29.73–44.70 days across the reported DARM cells, despite competitive short-horizon losses. Parameter uncertainty and forecast-functional uncertainty must therefore be separately reported.

13.2 Misspecification and the information set

In Experiments 16–18, DARM performed competitively for the tested ARCH-family generating laws. Experiment 17 introduced return-filtered SV and daily-squared-return HAR; its filtered-SV benchmark knew the generating Student-t parameter, unlike fitted DARM and GJR. Experiment 18 estimated ν in all three model families, though filtered SV used log-squared-return quasi-likelihood rather than the conditional Student-t likelihood employed by the ARCH-family estimators. Under baseline log-SV ($T = 4000, H = 10$), estimated-ν *expected oracle-relative QLIKE regret* was 0.002382 for filtered SV, 0.011279 for GJR-GARCH and 0.011543 for asymmetric DARM. These are not realised paired QLIKE improvements. In the same Experiment 18 cell the realised SV-versus-GJR improvement was 0.000127, with unadjusted 95% CI $[-0.007021, 0.007275]$. Under stress log-SV the analogous paired improvement was 0.048695 (CI $[0.014531, 0.082860]$). Thus neither ranking nor a selected unadjusted interval establishes a universal claim.

The log-SV *stable* regime supplies an instructive counterexample to a blanket SV-benchmark ranking: at $T = 4000, H = 10$ its expected regret was 0.000867 for asymmetric DARM and 0.002868 for filtered SV. Regimes vary multiple generating parameters jointly, so differences cannot be assigned to one cause. The Experiment 18 filtered-SV ν fit reached its numerical upper limit near 40 in 139/480 estimated-mode fits; this is a diagnostic of possible weak quasi-likelihood identification, not a proof that ν is mathematically unidentified.

14 Predictive-distribution calibration

The equality in Proposition 10.1 concerns conditional *means*, not predictive laws. Experiment 18 fitted-parameter predictive intervals use 199 simulated paths per origin and were evaluated at 12 dependent origins within each of 20 independent datasets per configuration. Statistical summaries first aggregate within datasets and then across independent datasets. The fitting uncertainty is omitted. The reported symmetric Student-t coverage intervals can extend beyond the admissible probability range and are *not* bounded-binomial confidence intervals.

In baseline log-SV, asymmetric DARM's RV coverage of 93.75% contrasts with only 50.83% for IV; filtered SV reports 92.08% and 91.25%. Asymmetric DARM's 19.58% and 45.42% IV coverage in stable and stress regimes respectively show that the mismatch is not confined to the baseline. Different predictive widths and centring can both contribute; the experiment does not identify a single causal

Table 3: Experiment 18, log-SV generating law, $T = 4000$, $H = 10$: empirical coverage of nominal 90% intervals. Each figure aggregates 20 independent datasets; origin-level intervals are dependent.

Regime	Forecasting model	RV coverage	IV coverage	Median IV width
Stable	Asymmetric DARM	.8625	.1958	.000480
Stable	Filtered SV	.8625	.7958	.002888
Baseline	Asymmetric DARM	.9375	.5083	.002729
Baseline	GJR-GARCH	.9208	.4750	.002661
Baseline	Filtered SV	.9208	.9125	.005577
Stress	Asymmetric DARM	.8583	.4542	.001271
Stress	Filtered SV	.8833	.9083	.003099

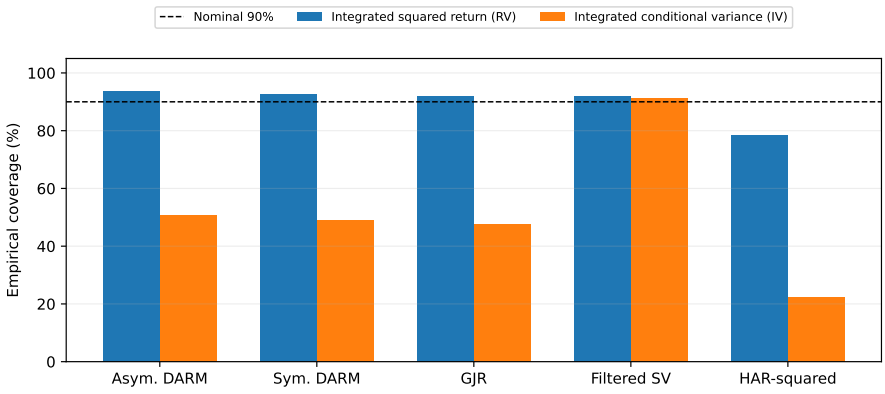


Figure 3: Experiment 18 baseline log-SV: 90% predictive coverage for realised squared returns and latent integrated conditional variance. Calibration for one target does not imply calibration for the other.

mechanism. The simulation cannot establish real-market calibration of latent IV, which is not directly observed in ordinary equity return panels.

15 Discussion, limitations and literature positioning

The finite DARM class is constrained ARCH, and infinite geometric DARM is exactly the finite-mean GARCH subclass described by the parameter map. The Volterra solution, covariance summability and companion-matrix methodology use established ARCH/random-recursion principles (Giraitis et al. 2000; Bougerol and Picard 1992; Nelson 1990). Parameterisation-specific deductions include explicit power-law normalisation sensitivities, ideal-weight identification from adjacent ratios, the renewal-pole persistence equation and its interpretation, and the associated kernel metric. Their originality relative to the entire parametric-ARCH literature has not been established by an exhaustive priority search.

The independent equity evaluation uses daily squared returns as noisy variance proxies, annual expanding re-estimation, correlated assets, overlapping horizons and a frozen script, but lacks the original version-pinned adjusted-price panel and fitted-parameter checkpoints. A historical point-in-time constituent universe has not been demonstrated. The headline p-values were audited against reported summaries and code but not independently regenerated from the unavailable origin-level market forecasts. In the synthetic studies, source notebooks and Python implementations for Experiments 13–18 are now archived, together with aggregate and (for Experiments 16–18) replication-level outputs; complete original simulation paths, oracle/filter trajectories, optimiser traces, and exact runtime random streams remain missing. Archive source availability does not certify exact historical execution-version equivalence. Table and confidence-interval arithmetic can be checked without certifying every forecast's nonanticipativity. The claimed primary endpoint is documented in the experiment script, not established by an independently dated preregistration. Bootstrap selected subsamples, multiplicity, and competing likelihood objectives further limit generality.

Statistically favourable mean forecast ranks are not equivalent to evidence of trading profitability, and the original independent ten-day contrast was inconclusive. Point-forecast accuracy does not identify a unique volatility mechanism; coverage of future squared-return intervals cannot substitute for credible latent-variance uncertainty. These distinctions motivate releases of complete execution environments and uncertainty diagnostics for kernel functionals, rather than stronger forecasting-superiority claims.

Data and Code Availability

The supporting source-code and reported-results collection is the separate *DARM Experiments 13–18 Reproducibility Archive*, file `DARM_Experiments_13_to_18_Reproducibility_Archive.zip` (assembled 19 October 2026). It contains authored Python source and notebooks for Experiments 13–17, the Experiment 18 authored notebook and a labelled verbatim code-cell extraction, executed-notebook outputs for Experiments 13–15, replication-level score tables for Experiments 16–18, aggregate reports, audit scripts, a manifest, and a SHA-256 inventory. An arithmetic/integrity validation script in that archive passed in this release audit; the original experiments were *not* independently rerun. Experiments 13–15 do not have separate original per-replication fitting records. No experiment has complete original simulation paths and optimizer histories; exact matches between archived source and historical Colab execution environments cannot be guaranteed. The financial-market study has an archived forecasting implementation and summary results, but the original adjusted equity-price panel and per-origin market forecast/loss records are unavailable. Accordingly, this is a source-code and reported-results release, *not* a claim of end-to-end computational repro-

ducibility. The accompanying technical supplement details the inventory and distinctions among original code, reported output and derived verification.

16 Conclusion

Normalised power-law DARM offers an interpretable low-dimensional restriction of ARCH memory with explicitly tractable persistence, moment and forecast representations. Its primary equity-market ten-day improvement over GJR-GARCH is not statistically established. Synthetic evidence shows practical weak identification despite strong short-horizon point forecasts and serious latent-integrated-variance interval undercoverage under some misspecified log-SV designs. The contribution is therefore a transparent model parameterisation, mathematical analysis of its lag geometry and a qualified empirical assessment of what its forecasts do—and do not—identify.

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